

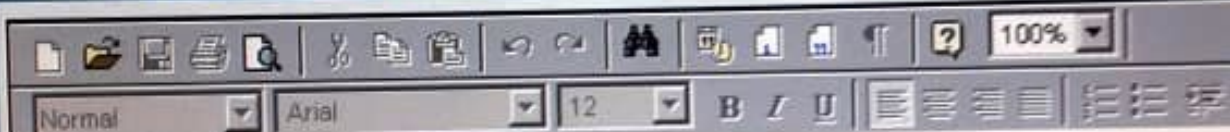
**MTH622** Vectors and Classical Mechanics

Question No : 39 of 41

Find the moments of inertia of a ring of radius $2a$ about an axis through center.

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Answer ([Please click here to Add Answer](#))



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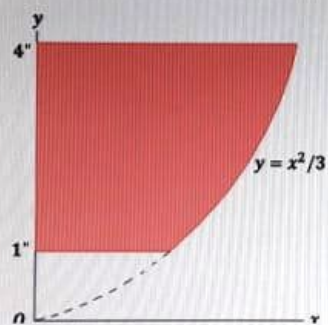
MTH622 Vectors and Classical Mechanics

MC18

Question No : 41 of 41

Marks: 5 (Budgeted Time 10 Min)

Calculate the moment of inertia of the shaded area given in figure about y-axis.



Answer ([Please click here to Add Answer](#))

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MTH622 Vectors and Classical Mechanics

MC180

Question No : 40 of 41

Marks: 5 (Budgeted Time 10 Min)

Calculate the moment of inertia of a uniform rod of length L about an axis perpendicular to the rod and passing through an end point.
(Let the x -axis be chosen along the length of the rod, with origin at one end point. Let M and a be the mass and length of rod respectively and suppose the rod to be composed of small elements.)

Answer (Please [click here to Add Answer](#))

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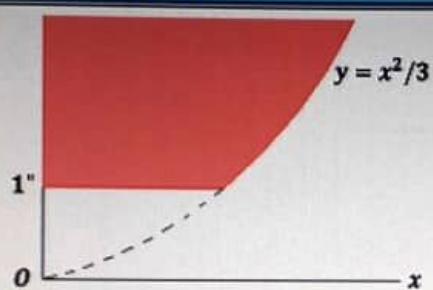
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MTH622 Vectors and Classical Mechanics

Question No : 41 of 41

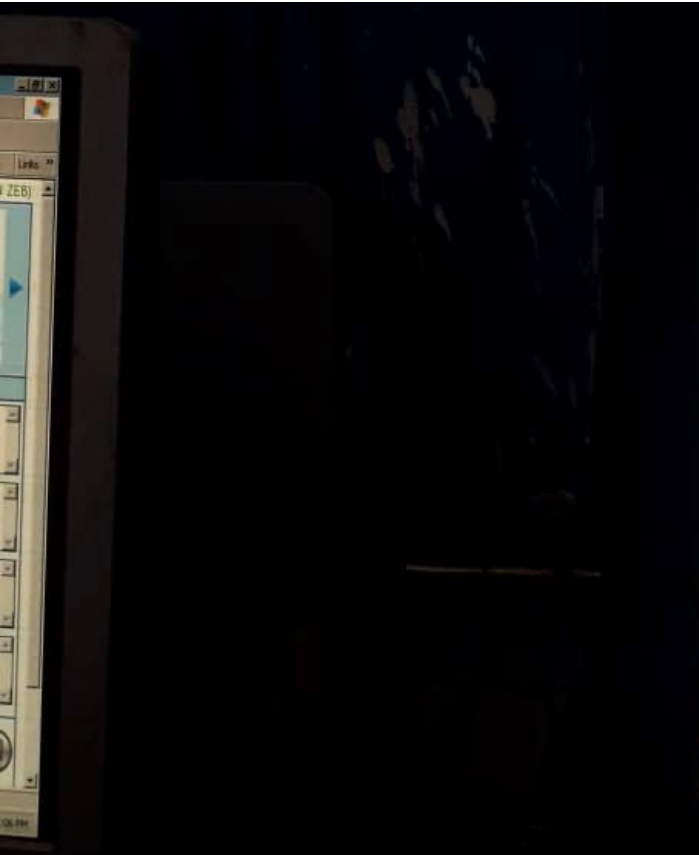
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whereas an elementary strip from the shaded area whose Moment of inertia is

Answer (Please [click here](#) to Add Answer)





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MTH622 Vectors and Classical Mechanics MC180403

Question No: 36 of 41 Marks: 3 (Budgeted Time: 6 Min)

Consider a small element of a uniform elliptic plate has moment of inertia $I = \delta m y^2$. Write down the expression for moment of inertia of whole plate.
(Note: Do not solve the expression)

Answer: [Please click here to Add Answer](#)

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MTH622 Vectors and Classical Mechanics **MC1804**

Question No: 37 of 41 Marks: 3 (Budgeted Time 6 Min)

If a coordinate system rotates with angular velocity $\vec{\omega} = 2\hat{i} - 3\hat{j} + 5\hat{k}$ relative to the fixed coordinate system, and $\vec{A} = \sin t \hat{i} - \cos t \hat{j} + e^{-t} \hat{k}$ then find $d\vec{A}/dt$ with respect to the fixed point system.

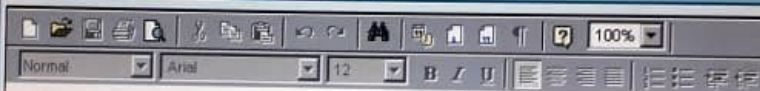
Where $\left(\frac{d\vec{A}}{dt}\right)_r = -\sin t \hat{i} + \cos t \hat{j} - 2e^{-t} \hat{k}$ and $\vec{\omega} \times \vec{A} = (4e^{-2t} - 3 \cos t)\hat{i} - (-2e^{-t} + 3 \sin t)\hat{j} + (\cos t + 2 \sin t)\hat{k}$

Answer ([Please click here to Add Answer](#))

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What is the impulse developed by a mass of 100 gm which changes its velocity from $i+j+k$ to $2i+2j+3k$ m/sec?

Answer ([Please click here to Add Answer](#))



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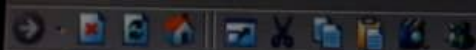
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MTH622 Vectors and Classical Mechanics

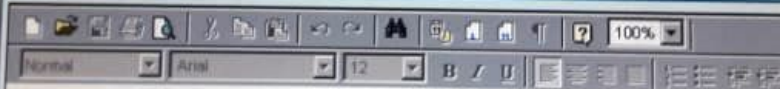
MC180403

Question No : 32 of 41

Marks: 2 (Budgeted Time 4 Min)

To find the inertia matrix of uniform solid cuboid (parallelepiped) at one of its corner, if $I_{xx} = \frac{M}{3}(b^2 + c^2)$
Then find I_{yy} , and I_{zz} .

Answer ([Please click here to Add Answer](#))



If moment of inertia of a uniform circular cylinder of length b and radius a about x-axis through the center and perpendicular to the central axis is:

$$I_x = m \left(\frac{1}{4} a^2 + \frac{1}{12} b^2 \right)$$

Answer ([Please click here to Add Answer](#))

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MTH622 Vectors and Classical Mechanics

MC

Question No : 34 of 41

Marks: 2 (Budgeted Time 4 Min)

Write the expression for total energy of a particle executing Simple Harmonic Motion.

Answer ([Please click here to Add Answer](#))

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MTH622 Vectors and Classical Mechanics

MC1804

Question No : 35 of 41

Marks: 3 (Budgeted Time 5 Min)

Two objects push off against one another starting from a stationary position. First object with mass 65 kg acquires a velocity of 2 m/s to the right. What velocity does the second object having mass 70 kg acquire?

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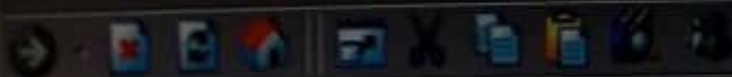
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**MTH622** Vectors and Classical Mechanics

Question No : 28 of 41

If there is no net force acting on body, then its acceleration is

Answer (Please select your correct option)

☒ zero☐☐ constant☐☐ increasing☐☐ decreasing☐

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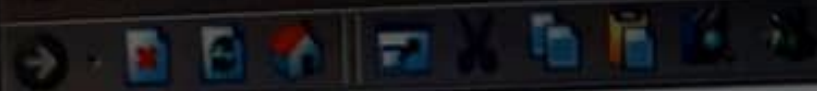
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Address <http://server/vutes/client/Paper.aspx>**MTH622** Vectors and Classical Mechanics

Question No : 29 of 41

Total amount of mass and energy together in a system is

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Answer (Please select your correct option)

☒ increasing☐ decreasing☐ zero☐ constant

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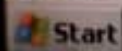
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MTH622 Vectors and Classical Mechanics

Question No : 30 of 41

Measure of average kinetic energy of molecules is

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Answer (Please select your correct option)

☒ temperature

☐

☐ energy

☐

☐ internal energy

☐

☐ enthalpy

☐

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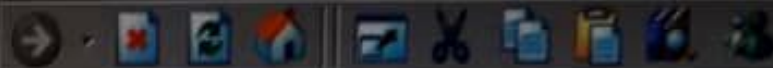


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MTH622 Vectors and Classical Mechanics

Question No : 31 of 41

What is a degrees of freedom of the system of rigid bodies.

Answer ([Please click here to Add Answer](#))



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MTH622 Vectors and Classical Mechanics

Question No : 27 of 41

Until a force acts on a body, it's velocity is

Answer (Please select your correct option)

☐ zero

☐ constant

☒ increasing

☐ decreasing

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MTH622 Vectors and Classical Mechanics

Question No : 26 of 41

The time taken by the particle to complete one oscillation is called

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Answer (Please select your correct option)

☐ amplitude

☒ time period

☐ frequency

☐ none of these

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MTH622 Vectors and Classical Mechanics

Question No : 25 of 41

The path of projectile is called its

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Answer (Please select your correct option)

☐ velocity

☐ acceleration

☒ trajectory

☐ none of these

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In a conservative field, the total energy of particle _____ throughout the motion.

Answer (Please select your correct option)

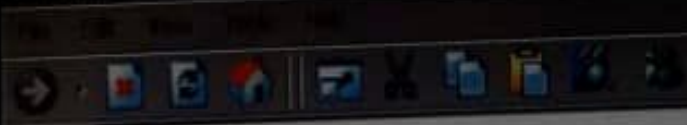
☐ varies rapidly

☐ does not remain constant

☒ remains constant

☐ none of these

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MTH622 Vectors and Classical Mechanics

Question No : 20 of 41

Work done is said to be zero, when

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Answer (Please select your correct option)

- ☐ a. Some force acts on a body, but displacement is zero
- ☐ b. No force acts on a body but some displacement takes place
- ☐ c. Either a) or b)
- ☐ d. None of these

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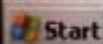
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MTH622 Vectors and Classical Mechanics

Question No : 21 of 41

The rate of change of displacement of a body is called

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Answer (Please select your correct option)

☒ Velocity

☐ Acceleration

☐ Momentum

☐ None of these

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MTH622 Vectors and Classical Mechanics

Question No : 22 of 41

For an irrotational flow

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Answer (Please select your correct option)

☒ $\nabla \times \vec{v} = \vec{0}$

☐ $\nabla \times \vec{v} \neq \vec{0}$

☐ $\nabla \cdot \vec{v} = \vec{0}$

☐ None of these

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MTH622 Vectors and Classical Mechanics

Question No : 23 of 41

The path of a projectile in a non-resistive medium is

Answer (Please select your correct option)

- ☐ linear
- ☐ Circular
- ☒ Parabolic
- ☐ Hyperbolic

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Question No : 19 of 41

Marks: 1 (Budgeted Time 1 Min)

The periodic time of a particle with simple harmonic motion is _____ proportional to the angular velocity

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Answer (Please select your correct option)

☐ Directly

☒ Inversely

☐ Square root

☐ None of these

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Question No : 18 of 41

Marks: 1 (Budgeted Time 1 Min)

For a conservative holonomic dynamical system, the Lagrangian L , kinetic energy T and potential energy V are connected by

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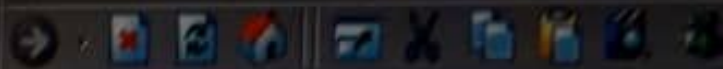
Answer (Please select your correct option)

☒ $L = T + V$

☐ $L = T - V$

☐ $L = 2T + V$

☐ $L = 2T - V$

**MTH622** Vectors and Classical Mechanics

Question No : 17 of 41

At pivot point, body with mass ' m ' will have a torque by the

Answer (Please select your correct option)

☒ force☐☐ circle☐☐ central point☐☐ radius of circle

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MTH622 Vectors and Classical Mechanics MC11

Question No : 16 of 41 Marks: 1 (Budgeted Time: 1 Min)

According to parallel axis theorem, the moment of inertia of a section about an axis parallel to the axis through center of gravity (i.e. IP) is given by _____
(where, A = Area of the section, IG = Moment of inertia of the section about an axis passing through its C.G., and h = Distance between C.G. and the parallel axis.)

Answer (Please select your correct option)

☒ $IP = IG + Ah^2$

☐ $IP = IG - Ah^2$

☐ $IP = IG / Ah^2$

☐ $IP = Ah^2 / IG$

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MTH622 Vectors and Classical Mechanics

Question No : 13 of 41

Cylinder has moment of inertia given by formula of

Answer (Please select your correct option)

☐ (ml^2)

☐ mr

☐ $\frac{1}{2}(mr)$

☐ $5(mr^2)$

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MTH622 Vectors and Classical Mechanics

Question No : 14 of 41

Moment of inertia of sphere is given as

Answer (Please select your correct option)

☐ $2(ml^2)$

☐ r^2

☐ $1/2(r)$

☒ $2/5(mr^2)$

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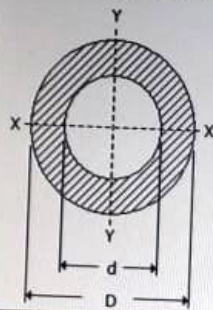


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Question No : 15 of 41

Marks: 1 (Budgeted Time 1 Min)

Moment of inertia of a hollow circular section, as shown in the below figure about X-axis, is



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Answer (Please select your correct option)

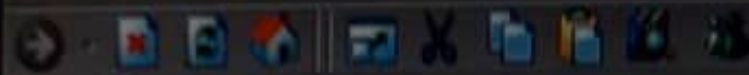
☐ $\pi/16 (D^2 - d^2)$

☐ $\pi/16 (D^3 - d^3)$

☐ $\pi/16 (D^4 - d^4)$

☒ $\pi/64 (D^4 - d^4)$

Done

**MTH622** Vectors and Classical Mechanics

Question No : 12 of 41

Moment of inertia of thin rod is given by formula of

Answer (Please select your correct option)

☐ $\frac{1}{12}(ml^2)$



☐ mr^2



☐ $2(mr)$



☐ $\frac{2}{5}(mr^2)$



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MTH622 Vectors and Classical Mechanics

MC180

Question No : 11 of 41

Marks: 1 (Budgeted Time 1 Min)

The moment of inertia of a circular section of base 'b' and height 'h' about an axis passing through its vertex and parallel to base is

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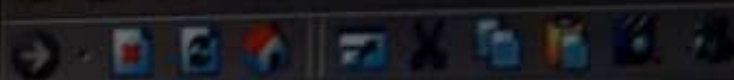
Answer (Please select your correct option)

☒ $\frac{bh^3}{12}$

☐ $\frac{bh^3}{36}$

☐ $\frac{bh^3}{3}$

☐ $\frac{bh^3}{4}$

**MTH622** Vectors and Classical Mechanics

Question No : 10 of 41

The wheels of a moving car possess

Answer (Please select your correct option)

☐ Potential energy only☐ Kinetic energy of translation only☐ Kinetic energy of rotation only☒ Kinetic energy of translation and rotation both

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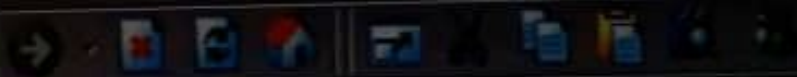


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Question No : 9 of 41

The polar moment of inertia of a circular section is about

Answer (Please select your correct option)

☐ X-X axis☐ Y-Y axis☒ Z-Z axis☐ Neutral axis

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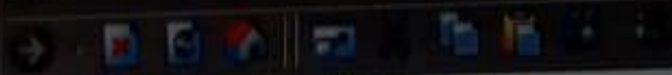
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MTH622 Vectors and Classical Mechanics

Question No : 5 of 41

The functions in Green's theorem will be

Answer (Please select your correct option)

☒ Continuous derivatives

☐

☐ Discrete derivatives

☐

☐ Continuous partial derivatives

☐

☐ Discrete partial derivatives

☐

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MTH622 Vectors and Classical Mechanics

MC18

Question No : 6 of 41

Marks: 1 (Budgeted Time 1 Min)

If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fixed and rotating coordinate system, then

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Answer (Please select your correct option)

☐ $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - \omega \times \vec{A}_r$

☒ $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r$

☐ $\left(\frac{d\vec{A}}{dt}\right)_f = A_r - \omega \times \vec{A}_r$

☐ $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - A \times \vec{A}_r$

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MTH622 Vectors and Classical Mechanics

Question No : 7 of 41

The axes relative to which product of inertia are zero are called

Answer (Please select your correct option)

- ☒ principal axes
- ☐ principal moment of inertia
- ☐ moment of inertia
- ☐ axes of rotation

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A 3×3 symmetrical matrix has ----- real eigenvalues, which may be distinct or repeated

Answer (Please select your correct option)

☐ Infinite

☐ Zero

☐ Two

☒ Three

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MTH622 Vectors and Classical Mechanics

Question No : 4 of 41

The Stoke's theorem can be used to find which of the following?

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Answer (Please select your correct option)

- ☒ Area enclosed by a function in the given region
- ☐ Volume enclosed by a function in the given region
- ☐ Linear distance
- ☐ Curl of the function

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MTH622 Vectors and Classical Mechanics

Question No : 3 of 41

$$\nabla \times \nabla \varphi = ?$$

Answer (Please select your correct option)

☐ $\nabla^2 \varphi$

☐ φ

☒ 0

☐ None of these

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Which of the following property is not true?

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Answer (Please select your correct option)

☐ $\nabla \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$

☐ $\nabla \cdot (\varphi \vec{A}) = \varphi (\nabla \cdot \vec{A}) + \vec{A} \cdot (\nabla \varphi)$

☐ $\nabla \cdot \nabla \varphi = \nabla^2 \varphi$

☒ $\nabla \cdot \vec{V} = \vec{V} \cdot \nabla$

The maximum value of the directional derivative of $\varphi(x, y, z)$ is equal to the

Answer (Please select your correct option)

☐ Magnitude of $\nabla \times \varphi$

☐ Magnitude of φ

☒ Magnitude of $\nabla \varphi$

☐ None of these